

LastFM Dataset Results

(a) RS domain					(b) IR domain				
Approach	@k	NDCG	Precision	Recall	Approaches	@k	NDCG	Precision	Recall
RS models					IR models				
LightFM	5	0.6455	0.9541	0.3805	BM25	5	0.0194	0.0159	0.0023 [†]
	10	0.7184	0.8765	0.6978		10	0.0157	0.0109	0.0028
	20	0.7627	0.5603 [†]	0.8897		20	0.0122	0.0103	0.0043
	50	0.7801 [†]	0.2357	0.9356		50	0.0109	0.0078	0.0078 [†]
DeepFM	5	0.5677	0.8641	0.3444	Dense Retriever	5	0.0597	0.0479	0.0035 [†]
	10	0.6400	0.7902	0.6291		10	0.0502	0.0400	0.0056
	20	0.7206	0.5550 [†]	0.8818		20	0.0424	0.0333	0.0082
	50	0.7537 [†]	0.2439	0.9686		50	0.0363	0.0272	0.0155 [†]
BERT4Rec	5	0.6205 [†]	0.9296 [†]	0.3707	Bi-Encoders	5	0.0245	0.0240	0.0042 [†]
	10	0.7029	0.8676	0.6910		10	0.0216	0.0191	0.0057
	20	0.7604	0.5707	0.9065		20	0.0189	0.0147	0.0092
	50	0.7821	0.2431	0.9652		50	0.0191	0.0121	0.0165 [†]
LightGCN	5	0.6313	0.8598	0.4359	KNN-SPLADE	5	0.0159	0.0137	0.0039
	10	0.6928	0.7344	0.7336		10	0.0139	0.0107	0.0051
	20	0.7576	0.4622	0.9098		20	0.0142	0.0108	0.0074
	50	0.7812	0.1989	0.9763		50	0.0152	0.0097	0.0138
ItemKNN	5	0.4353	0.6323	0.3643	ColBERT	5	0.1156	0.1087	0.0122
	10	0.6000	0.6744	0.7675		10	0.1084	0.0975	0.0196
	20	0.7303	0.4857	0.9313		20	0.0987	0.0864	0.0317
	50	0.7468	0.2107	0.9701		50	0.0912	0.0727	0.0559
IR -> RS mapping					RS -> IR mapping				
Centroid Vector ^a	5	0.1181	0.1155	0.0733	Query-as-User ^c	5	0.0669	0.0511	0.0335
	10	0.1284	0.1063	0.1348		10	0.0667	0.0422	0.0463 [†]
	20	0.1810	0.0944	0.2392		20	0.0699	0.0336	0.0737 [†]
	50	0.2793	0.0771	0.4879		50	0.0795	0.0256	0.1168
Tag Query ^b	5	0.6100	0.5436	0.3425	Retrieval-as-User ^c	5	0.1039	0.1033	0.0277 [†]
	10	0.4862	0.3060	0.3855		10	0.0972	0.0933	0.0492
	20	0.5018	0.1652	0.4165		20	0.0903	0.0770	0.0802
	50	0.5383	0.0808	0.5097		50	0.1054	0.0443	0.1142 [†]

Table 1: LastFM: BR2IDGE top-k mean results. **Bold** marks best values per @k/metric; [†] indicates statistical equivalence (t-test, p-mode=holm, $\alpha = 0.05$). Hybrid backbones: ^a Centroid Vector uses *Bi-Encoder*; ^b Tag Query uses *BM25*; ^c RS $\rightarrow$ IR strategies use *LightFM*.